Kaustubh Parag Khachane

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.39
Sant Mai Junior College, Shivaji Nagar, Pune	Class XII (Maharashtra State Board of Secondary and Higher Education)	2023	86.50%
Tree House High School, Karvenagar	Class X (ICSE)	2021	94.16%

EXPERIENCE

Research Intern -- *Next Generation Wireless Systems Lab, Indian Institute of Science, Bangalore (May 2026 - July 2026)*

Efficient Channel Estimation for Reconfigurable Intelligent Surface (RIS)-Assisted Near-Field Communication (Advisor: *Prof. Neelesh B. Mehta***)**

- **Extended** RIS-assisted near-field channel estimation to **practical** wideband OFDM systems with multi-cluster scattering for Extremely Large Aperture Arrays, targeting **6G** communication systems
- **Formulated** the parametric channel estimation problem to jointly exploit pilot observations across all subcarriers, enabling a **tunable pilot overhead-accuracy tradeoff** via a configurable decimation factor
- **Implemented** and compared **EM**, **SAGE**, and **Bayesian Optimization** solvers, reducing required objective function evaluations by over an order of magnitude versus a **grid-based solver baseline** (~100,000 to ~1,000 evaluations). **Contributed to ongoing research paper**

Research Project *(May 2025 - Present)*

Receiver design and Performance analysis for OTFS Modulation and Channel Modelling ***(Guide:*** *Prof. Sundaram Vanka****)***

- Implemented an **SSM**-based framework in Python for **RSSI prediction** and used it to analyze how **predictability varies with the channel coherence time**, showing that **multi-step prediction degrades** consistently with Doppler-induced decorrelation and the **loss of phase information** in magnitude only observations.
- Implemented and compared **low-complexity LMMSE and MRC** equalizers from existing literature. Implemented and studied **Sinc**, **RRC**, and **Gaussian-Sinc** pulse shaping for **OTFS**, analyzing BER performance and matching the results with the literature.

PROJECTS

AI-Assisted Wi-Fi Radio Resource Management - *Inter IIT Tech Meet 14.0 (Oct 2025 - Dec 2025)*

- **Implemented** a CNN in **PyTorch** (4.8M parameters) with convolutional layers (32, 64, and 128 filters), batch normalization, max-pooling, and dropout to classify 18,750-point RF spectra into six interference classes, then integrated a discounted-UCB **multi-armed bandit (MAB)** controller to adapt spectrum-sensing dwell **time up to 120 ms from a 40 ms baseline.**
- **Deployed** the CNN using **TorchScript and C++ LibTorch** within the ns-3 Wi-Fi simulation, loading and warming it up once at startup and sharing it across all access points. Implemented **configuration guardrails** for channel switching, transmit power, and OBSS-PD adjustments.
- **Evaluated** a **10-AP** Wi-Fi network over **72 simulated hours** with **108** injected BLE, ZigBee, and microwave interference events, logging **12.4K control decisions** while sustaining **27 Mbps** average throughput.

Convex Optimization for Two-Layer ReLU Networks - *Course Project EE5606 Convex Optimization (Mar 2026 - Apr 2026)*

- Reimplemented **Mert Pilanci's ICML 2020** convex optimization framework, reformulating **non-convex** two-layer ReLU network training as an **equivalent convex** problem
- **Implemented** SGD-based ReLU training and the corresponding **CVXPY convex formulation** from scratch, comparing convergence and test accuracy on 1D, 2D, and binary **CIFAR-10 datasets**

- **Extended** with 3 heuristic sampling strategies to approximate the convex formulation at scale, and **applied Pilanci's CNN framework** to **benchmark** SGD-trained filters against a DFT-based convex baseline on CIFAR-10.

OFDM Channel Estimation on Software Defined Radio - *Communication Systems Lab (Aug 2025 - Dec 2025)*

- **Implemented** a QPSK-based **OFDM transceiver system** on **ADALM-Pluto** and **USRP B210** SDRs with pilot-assisted **channel estimation** in both single-device (**loopback**) and two-device (**Tx-Rx**) configurations.
- Successfully **demonstrated real time** OFDM transmission and channel estimation, verifying **detection**, **synchronization** and **BER** performance.

SKILLS

Programming: C/C++, Python, MATLAB, Verilog

Tools and Libraries: C++ STL, PyTorch , Scikit-learn, ADALM Pluto, Simulink, USRP B210

Software Fundamentals: OOPs, Data Structures and Algorithms, Complexity Analysis, Machine Learning,Optimization

POSITIONS OF RESPONSIBILITY

Teaching Assistant -- EE1030 Matrix Theory *(Aug 2024 - Dec 2024)*

- Organized quizzes, corrected assignments, and participated in doubt-solving sessions; clearly communicated technical concepts to a diverse cohort of students.

Teaching Assistant -- EE3700 Communication Systems *(Jul 2026 - present)*

- Prepared solution keys and corrected quizzes and assignments, ensuring correctness and clarity across Communication Systems concepts.

RELEVANT COURSES

Mathematics: Probability and Random Processes (A+), Matrix Theory, Complex Variables, Differential Equations, Introduction to Statistics (A+), Discrete Mathematics for Computer Science, Convex Optimization, Statistical Inference

Electronics and Communication: Signals and Systems, Control Systems, Communication Systems, Digital Signal Processing, DSP Lab, Wireless Communication, Information Theory

Computer Science: Introduction to Programming, Data Structures and Applications, Algorithms, Theory of Computation, Numerical Methods for Device Modelling , Computer Architecture, Operating Systems

Interdisciplinary: Fundamentals of Machine Learning, Deep Learning , Introduction to AI & ML, Communication Skills

SCHOLASTIC ACHIEVEMENTS

- **KVPY Fellow** (SA): All India Rank **751** in the National Science Aptitude Examination
- **Indian Olympiad Qualifier in Mathematics** (IOQM) 2021: Certificate of Merit (State Top 300); Qualified PRMO 2019
- **Inter-IIT Tech Meet 2025**: Part of the technical team in the IIT Hyderabad contingent; participated in a High-Prep event